BootIT

Ai Ting Lee

Based on slides by Niek Janssen
-- Based on slides by Sebastian Mateos Nicolajsen
---- Based on slides by Claus Brabrand

Welcome Back!!!

Agenda

- Recap
- Conditional Statements
  - If Statements
  - If-else Statements
  - Nested if(-else) Statements
- Booleans
- Returning Functions
- Boolean Comparisons
- Iterations / Loops
- Interactive Programs

Variables

Type Name Value

```
int x = 42;
System.out.println(x);
```

Value: What is actually stored?

Name: Identifier used to refer to this box.

`int` **Type:** Shape of the box; What can be put in?

Types

int

```
int x = 42;
System.out.println(x);
```

1, 2, 0, -2, -420, 1000000, 2147483647

String

```
String s = "hi";
System.out.println(s);
```

"hi", "hello world", "@#$%&*(),.;", "14"

double

```
double d = 3.14;
System.out.println(d);
```

1.5, 3.1415, -27.15, 1.0, 6.02E23

boolean

```
boolean b = true;
System.out.println(b);
```

true and false

Operations on Numbers

Operators in `int`

```
System.out.println(3 + 3);

int x = 3;
System.out.println(x + x);  6

x = 2;
System.out.println(x * x);  4

x = 6;
System.out.println(x / 3);  2
```

Operators in `double`

```
System.out.println(2.2 + 2.2);

double x = 2.2;
System.out.println(x + x);  4.4

x = 2.5;
System.out.println(x * 3);  7.5

x = 5.0;
System.out.println(x / 2);  2.5
```

Operations on Strings

+ is String Concatenation Operatort

```
String hi = "Hello ";
String world = "World!";
String greet = hi + world;

System.out.println(greet);
// Hello World!
```

It also works between Strings and numbers:

```
int year = 2026;
System.out.println("The year is " + year);
```

Methods

Methods let us **reuse code** and give a snippet a clear **responsibility**.

This does exactly the same as the previous exercise, wrapped in a method:

```
public class Main {
    static void dkkToEur() {
        double dkk = 100;
        double eur = dkk / 7.45;
        System.out.println(dkk + " kr corresponds to " + eur + " euro");
    }

    public static void main(String[] args) {
        dkkToEur();
    }
}
```

Methods with Parameters

But the previous method always converts 100 kr.

With a **parameter**, the same method works for any value:

```
public class Main {
    static void dkkToEur(double dkk) {
        double dkk = 100;
        double eur = dkk / 7.45;
        System.out.println(dkk + " kr corresponds to " + eur + " euro");
    }

    public static void main(String[] args) {
        dkkToEur();
        dkkToEur(100);
        dkkToEur(20);
    }
}
```

BootCode - Online IDE Built for BootIT

Again! Access the website via a laptop by...

Scanning the QR code

Entering the web address directly

https://cobblerscoders.tech/

Clicking the link on the course platform

ITU LEARNIT

https://learnit.itu.dk/course/view.php?id=3026730

Join today’s session, refresh if you don’t see it.

Agenda

- Recap
- **Conditional Statements**
  - If Statements
  - If-Else Statements
  - Nested if(-else) Statements
  - If-Else-If Ladder
- Booleans, Logical and Comparison Operators
- Returning Functions
- Iterations / Loops
- Interactive Programs

If Statements

- An **if** runs a block of code only when a condition is true. If the condition is false, Java simply skips the block and continues.

```
if (condition) {
    System.out.println("Yes the condition is correct");
}
```

Exercise - Conditional Printing


```
if (condition) {
    System.out.println("Yes the condition is correct");
}
```

You just walked into the ITU atrium. You have a boolean **atItu**.
Print "Welcome to ITU!" only if **atItu** is **true**.
If you set **atItu** to **false**, the program should print nothing.

Exercise - Conditional Printing


```
public class Main {
    public static void main(String[] args) {
        boolean atItu = true;
        if (atItu) {
            System.out.println("Welcome to ITU!");
        }
    }
}
```

Answer

If-Else Statements

- With **if-else**, exactly one of the two branches runs

```
// code that is always executed

if (condition) {
    System.out.println("To be");
} else {
    System.out.println("Not to be");
}

// code that is always executed
```

Exercise - Printing with If-Else


```
// code that is always executed

if (condition) {
    System.out.println("To be");
} else {
    System.out.println("Not to be");
}

// code that is always executed
```

Fun fact: Scrollbar is the Friday bar at ITU — open every Friday after 15:00 during the semester.

Write a method that prints whether it is Scrollbar day for a given weekday:

- static void **IsScrollBarOpen(String weekday)**
- **If weekday == "Friday"** → Yes, it is Friday, Scrollbar will open today!
- **Otherwise** → No, Scrollbar is closed.

Call it twice from main, once with weekday="Friday" and another with "Thursday"

Exercise - Printing with If-Else


```
public class Main {

    static void IsScrollBarOpen(String weekday) {
        if (weekday == "Friday") {
            System.out.println("Yes, it is Friday, Scrollbar will open today!");
        } else {
            System.out.println("No, Scrollbar is closed.");
        }
    }

    public static void main(String[] args) {
        IsScrollBarOpen("Friday");
        IsScrollBarOpen("Thursday");
    }
}
```

Answer

Nested if Statements

You can put an if inside another if. The inner block only runs when the outer condition is also true.

```
int number = 42;

if (number > 0) {
    if (number > 100) {
        System.out.println("positive and big");
    } else {
        System.out.println("positive but small");
    }
} else {
    System.out.println("not positive");
}
```

Exercise - Printing with Nested Ifs


```java
int number = 42;

if (number > 0) {
    if (number > 100) {
        System.out.println("positive and big");
    } else {
        System.out.println("positive but small");
    }
} else {
    System.out.println("not positive");
}
```

The ITU canteen serves lunch from Monday to Friday, 11:00 to 14:00.

Declare **isCanteenOpen(boolean isWeekday, double hour)** so it prints whether the canteen is open, using **nested if**, three layers deep:

1. Outer layer — is it a weekday? Use the parameter **isWeekday**.
2. Middle layer (only checked when it is a weekday) — is **hour < 11.0**?
3. Inner layer (only checked when the middle layer is false) — is hour **>= 14.0**?

Exercise - Printing with Nested Ifs


```java
public class Main {

    static void isCanteenOpen(boolean isWeekday, double hour) {
        if (isWeekday) {
            if (hour < 11.0) {
                System.out.println("Close");
            } else {
                if (hour > 14.0) {
                    System.out.println("Close");
                } else {
                    System.out.println("Open");
                }
            }
        } else {
            System.out.println("Close");
        }
    }

    public static void main(String[] args) {
        isCanteenOpen(false, 12.0); //Close
        isCanteenOpen(true, 9.0);  //Close
        isCanteenOpen(true, 11.0);  //Open
        isCanteenOpen(true, 14.0);  //Open
    }
}
```

Answer

If-Else-If Ladder

You can chain multiple conditions using **else if**. In this chain, only the first (from top to bottom) matching condition will execute.

```
if (number >= 100) {
    System.out.println("3+ digits");
} else if (number >= 10) {
    System.out.println("2 digits");
} else {
    System.out.println("1 digit");
}
```

Exercise - Printing with an If-Else-If Ladder


```java
if (number >= 100) {
    System.out.println("3+ digits");
} else if (number >= 10) {
    System.out.println("2 digits");
} else {
    System.out.println("1 digit");
}
```

The ITU Canteen offers a late lunch discount on weekdays after 13:45.
Assuming it is already a weekday, **implement** the lunch-hour check using a single **if-else-if ladder** that includes the late lunch discount logic and **print**:

Use a **double** for the time of day (e.g. 11.0 = 11:00, 13.75 = 13:45):

- Earlier than 11.0 (< 11.0) → Too early. Lunch starts at 11:00.
- Between 11.0 and 13.75 (>= 11.0 and < 13.75) → Lunch is being served at full price.
- Between 13.75 and 14.0 (>= 13.75 and < 14.0)→ Lunch is being served with a late lunch discount!
- After 14.0 (>= 14) → Too late - lunch ended at 14:00.

Exercise - Printing with an If-Else-If Ladder


```java
public class Main {

    static void printLunchStatus(double time) {
        if (time >= 14.0) {
            System.out.println("Too late - lunch ended at 14:00.");
        } else if (time >= 13.75) {
            System.out.println("Lunch is being served with a late lunch discount!");
        } else if (time >= 11.0) {
            System.out.println("Lunch is being served at full price.");
        } else {
            System.out.println("Too early - lunch starts at 11:00.");
        }
    }

    public static void main(String[] args) {
        printLunchStatus(10.5);  // Too early - lunch starts at 11:00.
        printLunchStatus(12.0);  // Lunch is being served at full price.
        printLunchStatus(13.8); // Lunch is being served with a late lunch discount!
        printLunchStatus(14.5);  // Too late - lunch ended at 14:00.
    }
}
```

Answer

Agenda

- Recap
- Conditional Statements
  - If Statements
  - If-Else Statements
  - Nested if(-else) Statements
  - If-Else-If Ladder
- **Booleans, Logical and Comparison Operators**
- Returning Functions
- Iterations / Loops
- Interactive Programs

Boolean

- A Boolean variable can take only two values: **true** or **false**

```
if(true) {
    System.out.println("This will ALWAYS be printed");
}

if(false) {
    System.out.println("This will NEVER be printed");
}
```

- Once you have a boolean, you can branch on it directly.

```
boolean isRaining = true;

if (isRaining) {
    System.out.println("Study");
} else {
    System.out.println("Enjoy the sun");
}
```

Exercise - Branching with Booleans


```java
boolean isRaining = true;

if (isRaining) {
    System.out.println("Study");
} else {
    System.out.println("Enjoy the sun");
}
```

The ITU fitness room (located in the basement, beneath Scrollbar) requires a membership.

Declare **checkFitnessAccess(boolean hasMembership)** so it stores membership in a **boolean** parameter, then **branches on the boolean** directly.

Print “Accessed” if they have a membership; **otherwise**, print “Not allowed”.

Exercise - Branching with Booleans


```
public class Main {
    public static void main(String[] args) {
        boolean hasMembership = false;

        if (hasMembership) {
            System.out.println("Accessed");
        } else {
            System.out.println("Not allowed");
        }
    }
}
```

Answer

Logical Operators

==	1 == 1	is equal to	-> true
!=	1 != 2	is not equal to	-> true
<	1 < 2	is less than	-> true
>	2 > 1	is greater than	-> true
<=	1 <= 2	is less than or equal to	-> true
>=	2 >= 2	is greater than or equal to	-> true

Comparison Operators

!	!(1 > 2)	not -> true, because (1 > 2) is false
&&	(2 > 1) && (3 > 2)	and -> true, both sides are true
||	(2 < 1) || (2 == 2)	or -> true, at least one side is true

P	Q	!P	P && Q	P || Q
T	T	F	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	F

Exercise - Branching with Boolean Operators

`==`	1 == 1	is equal to	-> true
`!=`	1 != 2	is not equal to	-> true
`<`	1 < 2	is less than	-> true
`>`	2 > 1	is greater than	-> true
`<=`	1 <= 2	is less than or equal to	-> true
`>=`	2 >= 2	is greater than or equal to	-> true

`!`	!(1 > 2)	not -> true, because (1 > 2) is false
`&&`	(2 > 1) && (3 > 2)	and -> true, both sides are true
`||`	(2 < 1) || (2 == 2)	or -> true, at least one side is true

The ITU fitness room offers a free trial on the first Tuesday of every month.

Now, you need to update the ITU fitness room access logic.
Declare **checkFitnessAccess(boolean hasMembership, boolean isFreeTrialTuesday)**:
a student may enter if they have a membership or if it is free-trial Tuesday.

Print "Accessed"; otherwise, print "Not allowed".

Exercise - Branching with Boolean Operators

16

```
public class Main {
    public static void main(String[] args) {
        boolean hasMembership = false;
        boolean isFreeTrialTuesday = true;

        if (hasMembership || isFreeTrialTuesday) {
            System.out.println("Accessed");
        } else {
            System.out.println("Not allowed");
        }
    }
}
```

Answer

Agenda

- Recap
- Conditional Statements
  - If Statements
  - If-Else Statements
  - Nested if(-else) Statements
  - If-Else-If Ladder
- Booleans, Logical and Comparison Operators
- **Returning Functions**
- Iterations / Loops
- Interactive Programs

Returning Values from Methods

- Instead of **void** (no return values), a method can send a specific value back to the caller using the **return** keyword.
- The **data type** being returned must be declared in the method **signature** exactly where void used to be.

```
public class Main {
    public static int addNumbers(int a, int b) {
        // Returns the calculated result to the caller
        return a + b;
    }

    public static void main(String[] args) {
        int result = addNumbers(2, 3);
        System.out.println(result);
    }
}
```

Exercise - Implementing a Method with a Return Value


```java
public class Main {
    public static int addNumbers(int a, int b) {
        // Returns the calculated result to the caller
        return a + b;
    }

    public static void main(String[] args) {
        int result = addNumbers(2, 3);
        System.out.println(result);
    }
}
```

Glass boxes are group study rooms in the ITU building.
You can reserve them on the same day at the info desk.
Some of these rooms operate on a "first-come, first-served" basis.

Declare a method named **IsFirstComeFirstServe** that returns a boolean.
The complete list of these room numbers is: 2A03, 2A07, 3A03, 4A01, 4A03, 4A07, 5A03, 5A07.

Make sure the method returns the correct values for the examples called in the main function.

Exercise - Implementing a Method with a Return Value


```
public class Main {

    static boolean IsFirstComeFirstServe(String roomNumber) {
        return (roomNumber == "2A03" || roomNumber == "2A07" || roomNumber == "3A03" ||
                roomNumber == "4A01" || roomNumber == "4A03" || roomNumber == "4A07" ||
                roomNumber == "5A03" || roomNumber == "5A07");
    }

    public static void main(String[] args) {
        System.out.println(IsFirstComeFirstServe("2A03")); // true

        boolean result_2A05 = IsFirstComeFirstServe("2A05");
        System.out.println(result_2A05); // false
    }
}
```

Answer

Agenda

- Recap
- Conditional Statements
  - If Statements
  - If-Else Statements
  - Nested if(-else) Statements
  - If-Else-If Ladder
- Booleans, Logical and Comparison Operators
- Returning Functions
- **Iterations / Loops**
- Interactive Programs

While Loops

- A **loop** is code that is **executed** many times

```
System.out.println("I should be careful with AI code!");
System.out.println("I should be careful with AI code!");
System.out.println("I should be careful with AI code!");
System.out.println("I should be careful with AI code!");
System.out.println("I should be careful with AI code!");
... x10
```

```
public static void main(String[] args) {
    int i = 0; // initialization
    while ( i < 10 ) { // loop condition
        System.out.println("I should be careful with AI code!");
        i = i  + 1; // incrementation
    }
}
```

i starts at 0

condition i < 10

i runs from 0 to 9 (10 times).

While Loops

- A while loop keeps running as long as its condition evaluates to true.
- The loop condition can be any **boolean expression**—including a boolean variable or a method that returns a boolean!
- The loop checks the condition **before** every iteration. Once the condition becomes false, the loop immediately stops and skips the body.
- You **must** update the variables involved in your loop condition. Otherwise you may create an **infinite loop** that runs forever and crashes your program!
- **Bad** example:

```
while(true){
    System.out.println("This never ends.");
}
```

Exercise - While the Sun is Out


```java
int i = 0;                    // initialization
while (i < 10) {              // loop condition
    System.out.println("I should be careful with the code that the AI wrote for me!");
    i = i + 1;                // increment
}
```

In Copenhagen, the summer days are long.
We love to sit on the grass outside the ITU building until the sun goes down at 21:00.

Declare a method named **isSunOut** that takes an **int** for the **time** and **returns** a **boolean**.

Use a **while loop** in the main function starting at 14:00.
The loop should keep running as long as the sun is out, checking the status and **incrementing** the time by 1 each round.

Exercise - While the Sun is Out


```java
public class Main {

    static boolean isSunOut(int time) {
        if (time < 21) {
            System.out.println("The time is now " + time + ": The sun is still out, let's sit on the
grass!");
            return true;
        } else {
            System.out.println("The time is now " + time + ": The sun is not out, let's go home.");
            return false;
        }
    }

    public static void main(String[] args) {
        // initialization
        int time = 14;
        boolean sunIsOut = true;

        while (sunIsOut) { // loop condition
            sunIsOut = isSunOut(time);
            time = time + 1; // increment
        }
    }
}
```

```
The time is now 14: The sun is still out, let's sit on the grass!
The time is now 15: The sun is still out, let's sit on the grass!
The time is now 16: The sun is still out, let's sit on the grass!
The time is now 17: The sun is still out, let's sit on the grass!
The time is now 18: The sun is still out, let's sit on the grass!
The time is now 19: The sun is still out, let's sit on the grass!
The time is now 20: The sun is still out, let's sit on the grass!
The time is now 21: The sun is not out, let's go home.
```

Answer

While Loops Quiz

- It is crucial to understand when a loop will exit and how the increment works inside the loop body.
- Let’s take a look at the following code snippets and predict their exact output.
- An "infinite loop" is a loop never stops.

Exercise - While Loop Quiz 1


```
int i = 10;
while (i > 0) {
    System.out.println(i);
    i = i - 1;
}
```

Exercise - While Loop Quiz 1


```java
int i = 10;
while (i > 0) {
    System.out.println(i);
    i = i - 1;
}
```

```
10
9
8
7
6
5
4
3
2
1
```

Answer

Exercise - While Loop Quiz 2


```java
int i = 1;
while (i <= 10) {
    System.out.println(i);
    i = i + 2;
}
```

Exercise - While Loop Quiz 2


```java
int i = 1;
while (i <= 10) {
    System.out.println(i);
    i = i + 2;
}
```

```
1
3
5
7
9
```

Answer

Exercise - While Loop Quiz 3


```java
int i = 1;
while (i < 100) {
    System.out.println(i);
    i = i * 2;
}
```

Exercise - While Loop Quiz 3


```
int i = 1;
while (i < 100) {
    System.out.println(i);
    i = i * 2;
}
```

```
1
2
4
8
16
32
64
```

Answer

Exercise - While Loop Quiz 4


```
int i = 1;
while (i < 42) {
    System.out.println(i);
    i = i * i;
}
```

Exercise - While Loop Quiz 4


```
int i = 1;
while (i < 42) {
    System.out.println(i);
    i = i * i;
}
```

```
infinite loop
```

Answer

Exercise - While Loop Quiz 5


```
int i = 0;
while (i <= 15) {
    System.out.println(i);
    i = i + 3;
}
```

Exercise - While Loop Quiz 5


```
int i = 0;
while (i <= 15) {
    System.out.println(i);
    i = i + 3;
}
```

```
0
3
6
9
12
15
```

Answer

Exercise - While Loop Quiz 6


```
int i = 64;
while (i >= 2) {
    System.out.println(i);
    i = i / 2;
}
```

Exercise - While Loop Quiz 6


```
int i = 64;
while (i >= 2) {
    System.out.println(i);
    i = i / 2;
}
```

```
64
32
16
8
4
2
```

Answer

Exercise - While Loop


```java
public static void main(String[] args) {
    int i = 0; // initialization
    while ( i < 10 ) { // loop condition
        System.out.println("I should be careful with AI code!");
        i = i  + 1; // incrementation
    }
}
```

Cafe Analog has a mobile app where you can buy 5 tickets at once with a discount.

You are a heavy coffee drinker and want 50 tickets in one go — so you need a **loop** that buys a pack of **5 tickets**, **ten times**.

Using a **while loop**, print the running total after each pack: 5, 10, 15, ..., 50 (one number per line).

Exercise - While Loop


```
public class Main {
    public static void main(String[] args) {
        int ticketsPerPack = 5;
        int total = 0;
        int packs = 0;
        while (packs < 10) {
            total = total + ticketsPerPack;
            System.out.println(total);
            packs = packs + 1;
        }
    }
}
```

Answer

For Loops

- The **for loop** is a simpler way to do loops based on counting
- Any while loop can be a for loop, but for loops make the **counter** and **condition** clearer.

```
public static void main(String[] args) {

    int i = 0; // initialization
    while ( i < 10 ) { // loop condition
        System.out.println("I should be careful with AI code!");
        i = i  + 1; // incrementation
    }

    for ( int i = 0; i < 10; i = i + 1 ) {
        System.out.println("II should be careful with AI code!");
    }
}
```

i starts at 0 | condition i < 10 | i runs from 0 to 9 (10 times).

Exercise - For Loop


```
for ( int i = 0; i < 10; i = i + 1; ) {
    System.out.println("II should be careful with AI code!");
}
```

Cafe Analog has a mobile app where you can buy 5 tickets at once with a discount.

You are a heavy coffee drinker and want 50 tickets in one go — so you need a **loop** that buys a pack of **5 tickets**, **ten times**.

Rewrite the Cafe Analog tickets program. Same output (5 … 50), but use a **for loop** this time.

Exercise - For Loop


```
public class Main {
    public static void main(String[] args) {
        int ticketsPerPack = 5;
        int total = 0;
        for (int packs = 0; packs < 10; packs = packs + 1) {
            total = total + ticketsPerPack;
            System.out.println(total);
        }
    }
}
```

Answer

Exercise - For Loop Quiz 1


```java
for (int i = 10; i > 0; i = i - 2) {
    System.out.println(i);
}
```

Exercise - For Loop Quiz 1


```java
for (int i = 10; i > 0; i = i - 2) {
    System.out.println(i);
}
```

```
10
8
6
4
2
```

Answer

Exercise - For Loop Quiz 2


```java
for (int i = 1; i < 10; i = i + 3) {
    System.out.println(i);
}
```

Exercise - For Loop Quiz 2


```java
for (int i = 1; i < 10; i = i + 3) {
    System.out.println(i);
}
```

```
1
4
7
```

Answer

Exercise - For Loop Quiz 3


```java
for (int i = 1; i < 10; i = i * i) {
    System.out.println(i);
}
```

Exercise - For Loop Quiz 3


```
for (int i = 1; i < 10; i = i * i) {
    System.out.println(i);
}
```

```
infinite loop
```

Answer

Exercise - For Loop Quiz 4


```java
for (int i = 0; i <= 15; i = i + 3) {
    System.out.println(i);
}
```

Exercise - For Loop Quiz 4


```java
for (int i = 0; i <= 15; i = i + 3) {
    System.out.println(i);
}
```

```
0
3
6
9
12
15
```

Answer

Exercise - For Loop Quiz 5


```java
for (int i = 1; i <= 10000; i = i * 10) {
    System.out.println(i);
}
```

Exercise - For Loop Quiz 5


```java
for (int i = 1; i <= 10000; i = i * 10) {
    System.out.println(i);
}
```

```
1
10
100
1000
10000
```

Answer

Exercise - For Loop Quiz 6


```java
for (int i = 64; i >= 2; i = i / 2) {
    System.out.println(i);
}
```

Exercise - For Loop Quiz 6


```java
for (int i = 64; i >= 2; i = i / 2) {
    System.out.println(i);
}
```

```
64
32
16
8
4
2
```

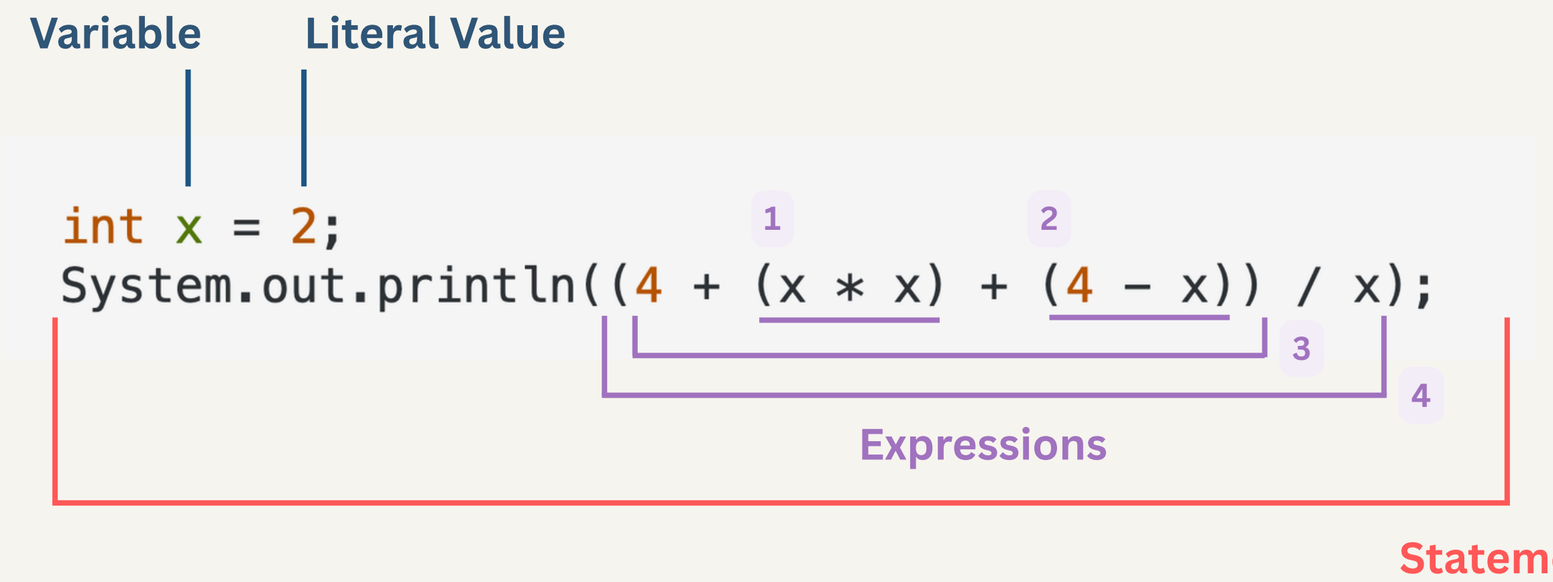
Answer

For Loop vs. While Loop

- **for loops** are more common than **while loops**
- They're more **compact**
- They're **easier** to read once you're used to it
- They are harder to mess up (it's really easy to forget the increment when making a while loop)
- The "**counter**" variables often have short names:
  - i, j, k
  - x, y, z

Recap: Anatomy of Java Code

- **Statement**: A complete unit of execution (usually ending with a semicolon ;).
- **Expression**: A code snippet that evaluates to a single value.
- **Literal Value**: A fixed value written directly in the source code.
- **Variable**: A named storage location of a specific type that holds a value.



Expressions

- **Constant (42)**
- **Variable (x)**
- **Comparison** between expressions
  - Exp == Exp
  - Exp != Exp
  - Exp > Exp
  - Exp < Exp
  - Exp >= Exp
  - Exp <= Exp
- **Operation** between expressions
  - Exp + Exp
  - Exp – Exp
  - Exp * Exp
  - Exp / Exp
  - Exp % Exp
- **Logical Operators**
  - Exp && Exp
  - ! Exp
  - Exp || Exp

Statements

- **Assignment**
  - Variable = Expressions
  - Variable = Constant

- **Conditional**
  - if(expression) { statement }
  - if(expression) { statement } else { statement }

- **Loops**
  - while(expression) { statement }
  - for(initialization, expression, iteration) { statement }

Nested for Loops

Sometimes you need two counters.

A nested for reads like the shape of the data:

```
public class Main {
    public static void main(String[] args) {
        for (int y = 0; y < 3; y++) {
            for (int x = 0; x < 3; x++) {
                System.out.print("(" + x + ", " + y + ") ");
            }
            System.out.println();
        }

    }
}
```

```
(0, 0) (1, 0) (2, 0)
(0, 1) (1, 1) (2, 1)
(0, 2) (1, 2) (2, 2)
```

Exercise - Using Nested for loops


```java
public class Main {
    public static void main(String[] args) {
        for (int y = 0; y < 3; y++) {
            for (int x = 0; x < 3; x++) {
                System.out.print("(" + x + ", " + y + ") ");
            }
            System.out.println();
        }

    }
}
```

You are in the ITU fitness room. Today's plan: 4 sets, 12 reps each.

Use a **nested for** to log every rep — outer loop = set (1…4), inner loop = rep (1…12).

The program should print :
Set 1 Rep 1
Set 1 Rep 2
Set 1 Rep 3
…
Set 4 Rep 12

Exercise - Using Nested for loops


```
public class Main {
    public static void main(String[] args) {
        for (int set = 1; set <= 4; set = set + 1) {
            for (int rep = 1; rep <= 12; rep = rep + 1) {
                System.out.println("Set " + set + " Rep " + rep);
            }
        }
    }
}
```

Answer

Exercise - More Loops Practice


```java
public static void main(String[] args) {

    int i = 0; // initialization
    while ( i < 10 ) { // loop condition
        System.out.println("I should be careful with AI code!");
        i = i  + 1; // incrementation
    }

    for ( int i = 0; i < 10; i = i + 1; ) {
        System.out.println("II should be careful with AI code!");
    }
}
```

Cafe Analog wants fewer disposable cups. Every time you buy a drink and bring your own cup, you get a stamp on your card — the 10th stamp is a free cup, and you start a fresh card right after.

Simulate 24 drinks bought (one per loop iteration) with a **for or while loop**:
stamps **starts at 0**, and each drink adds one stamp.

- While **stamps != 10:** print "Brought my own cup, got a stamp! ({stamps}/10)".
- The moment **stamps == 10**: print "Free cup! Here's a new stamp card." instead, then reset stamps back to 0.

Exercise - More Loops Practice


```
public class Main {
    public static void main(String[] args) {
        int drinksBought = 24;
        int stamps = 0;

        for (int drink = 1; drink <= drinksBought; drink = drink + 1) {
            stamps = stamps + 1;

            if (stamps != 10) {
                System.out.println("Brought my own cup, got a stamp! (" + stamps + "/10)");
            } else {
                System.out.println("Free cup! Here's a new stamp card.");
                stamps = 0;
            }
        }
    }
}
```

Answer

Agenda

- Recap
- Conditional Statements
  - If Statements
  - If-Else Statements
  - Nested if(-else) Statements
  - If-Else-If Ladder
- Booleans, Logical and Comparison Operators
- Returning Functions
- Iterations / Loops
- **Interactive Programs**

Random

Think of Java's **utilities** as a built-in toolbox. You can use these tools by **importing** them at the top of your program.

java.util.Random is one of them, which generates pseudo-random numbers.

For example, calling nextInt(4) produces 0, 1, 2, or 3 (each 25% likely).

```
import java.util.Random;

public class Main {
    public static void main(String[] args) {
        Random rng = new Random();
        int roll = rng.nextInt(4);
        if (roll == 0) {
            System.out.println("Zero");
        }
    }
}
```

Exercise - Using Random in a Loop

Friday at Scrollbar means one thing: beer pong! Let's set up the rack and simulate a full game.

Part 1: Use a **nested for loop** to print a 4-row triangle cup rack.

(Hint: Think about how the row number relates to the number of cups in that row).
Expected output for this part: Row 1: O Row 2: O O Row 3: O O O Row 4: O O O O

Part 2: Simulate throws using a **while loop** until all 10 cups are cleared.
You will need to keep track of the **remaining cups** and the **total number of throws**.

Game Rules:

- For each throw, use **java.util.Random** to generate a random number between 0 and 1 (inclusive).
- Treat a roll of 0 as a hit (a 50% chance). A hit removes one cup.
- A roll of 1 is a miss.
- Print the result of every throw. When the rack is empty, print a final game over message.

Expected output format for Part 2:
Throw 1: MISS! 10 cups left. Throw 2: SPLASH! 9 cups left. Throw 3: MISS! 9 cups left. ... Throw 17: SPLASH! 0 cups left. GAME OVER — the rack is empty. Chug up!

Exercise - Using Random in a Loop


```java
import java.util.Random;

public class Main {
    public static void main(String[] args) {
        for (int row = 1; row <= 4; row = row + 1) {
            System.out.print("Row " + row + ":");
            for (int cup = 1; cup <= row; cup = cup + 1) {
                System.out.print(" O");
            }
            System.out.println();
        }

        Random rng = new Random();
        int cupsLeft = 10;
        int throwNumber = 0;

        while (cupsLeft > 0) {
            throwNumber = throwNumber + 1;
            int roll = rng.nextInt(2);
            if (roll == 0) {
                cupsLeft = cupsLeft - 1;
                System.out.println("Throw " + throwNumber + ": SPLASH! " + cupsLeft + " cups left.");
            } else {
                System.out.println("Throw " + throwNumber + ": MISS! " + cupsLeft + " cups left.");
            }
        }

        System.out.println("GAME OVER - the rack is empty. Chug up!");
    }
}
```

Answer

Scanner (User Input)

Just like Random, Scanner is another essential tool you can grab from Java's built-in toolbox.

Scanner allows your program to **pause** and **listen** to the **user's keyboard input**.

By importing **java.util.Scanner** at the top of your program, you can ask questions, read the answers, and make your code truly interactive!

```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("Hello there!");
        System.out.print("What is your name? ");

        String userName = scanner.nextLine();

        System.out.println("Nice to meet you, " + userName + "!");

        scanner.close();
    }
}
```

Bonus Exercise - Using Scanner in VS Code

```
import java.util.*;
class Guess {
public static void main(String[] args) {
    Random random = new Random();
    Scanner scanner = new Scanner(System.in);
    int secret = random.nextInt(100);
    int tries = 0;
    int guess = -1;
    // As long as the correct number has not been guessed{
        // output make a guess
        guess = scanner.nextInt(); // user inputs guess
        // update tries
        // if guess is less than secret: print 'too low'
        // if guess is higher than secret: print 'too high'
    // }
    // print a congratulations message
    // print 'you used ... guesses'
    }
}
```

Bonus Exercise - Using Scanner in VS Code

```java
import java.util.*;
class Guess {
public static void main(String[] args) {
    Random random = new Random();
    Scanner scanner = new Scanner(System.in);
    int secret = random.nextInt(100);
    int tries = 0;
    int guess = -1;
    while (guess != secret){
        System.out.print("Guess a number between 0 and 99:");
        guess = scanner.nextInt();
        tries++;
        if(guess < secret){
            System.out.println("Too low");
        } else if(guess > secret) {
            System.out.println("Too high");
        }
    System.out.println("YOU WIN!");
    System.out.println("You used" + tries + " guesses.");
}
```

Answer